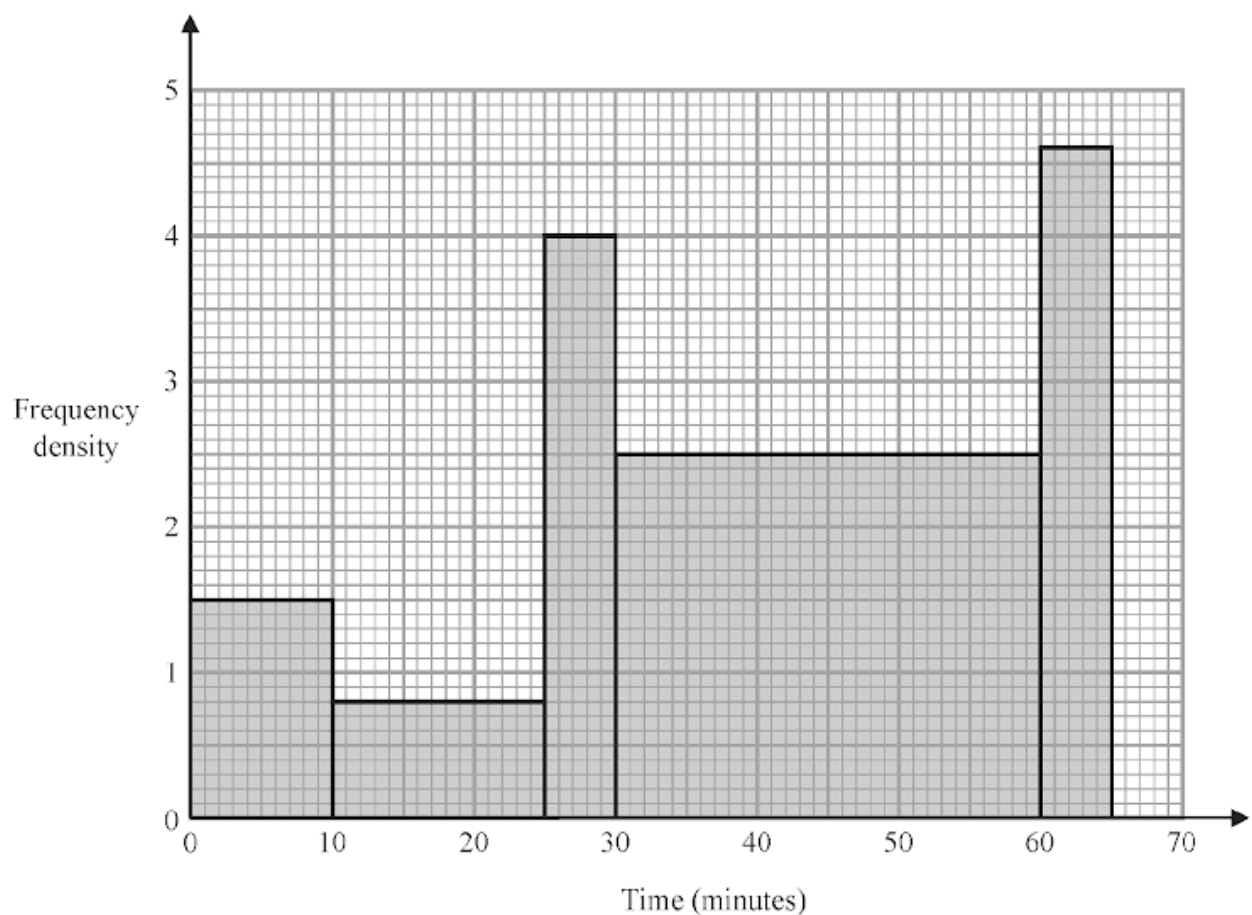


4MA1_1H_ch6_set_statistics_probability

Nguyen Khac Kiem PhD

1. 4MA1/1H_Winter_2024_Q18

The histogram shows information about the times some students took to complete a puzzle.



Work out an estimate for the fraction of these students who took between 20 minutes and 60 minutes to complete the puzzle.

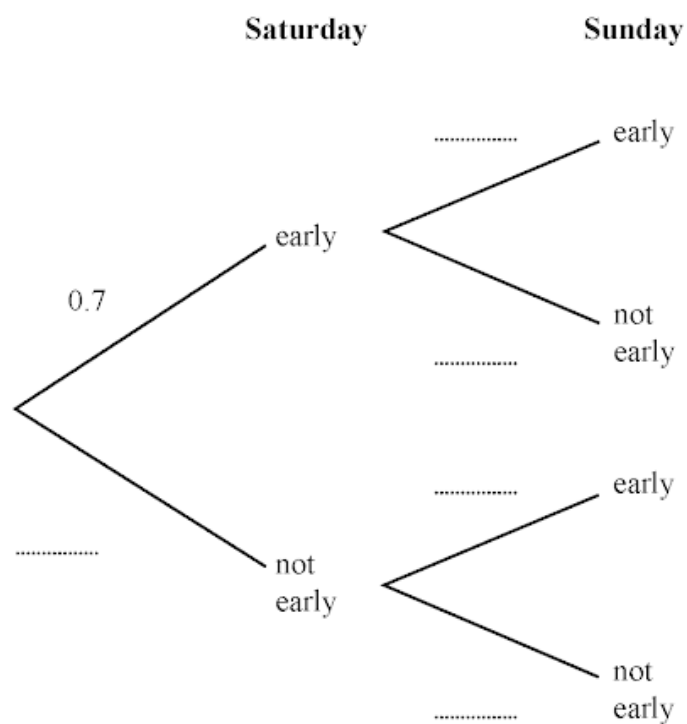
4 marks

The probability that Thomas gets to work early on Saturday is 0.7

If Thomas gets to work early on Saturday, the probability that he will get to work early on Sunday is 0.9

If Thomas does **not** get to work early on Saturday, the probability that he will get to work early on Sunday is 0.6

(a) Use this information to complete the probability tree diagram.



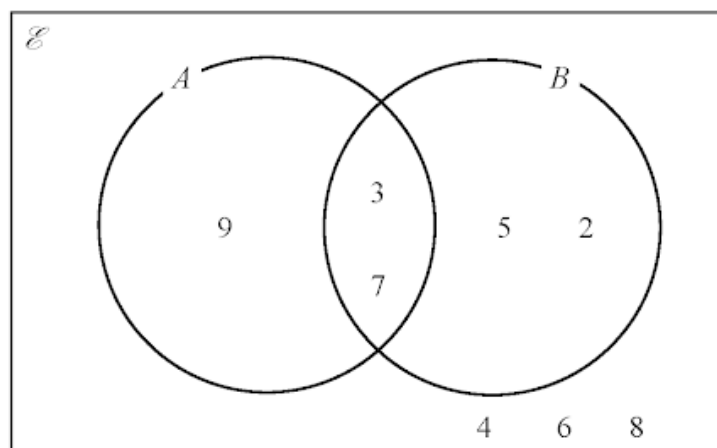
(2)

(b) Work out the probability that Thomas gets to work early on both Saturday and Sunday.

.....
(2)

3. 4MA1/1H_Winter_2024_Q4

Here is a Venn diagram.



(a) List the members of the set B

.....
(1)

(b) List the members of the set $A \cap B$

.....
(1)

(c) List the members of the set A'

.....
(1)

The table shows some information about the hourly rates of pay of 60 workers.

Hourly rate of pay (p dollars)	Frequency
$10 < p \leq 15$	18
$15 < p \leq 20$	16
$20 < p \leq 25$	14
$25 < p \leq 30$	8
$30 < p \leq 35$	4

(a) Write down the modal class.

.....
(1)

(b) Work out an estimate for the mean hourly rate of pay of the 60 workers.

..... dollars
(4)

5. 4MA1/1HR_Summer_2024_Q21

There are 25 counters in a bag such that

- 6 counters are blue
- x counters are orange, where $x > 9$
- the rest of the counters are pink

Maalam takes at random two of the counters from the bag.

The probability that Maalam takes one orange counter and one pink counter is $\frac{22}{75}$

Calculate the probability that Maalam takes 2 pink counters from the bag.

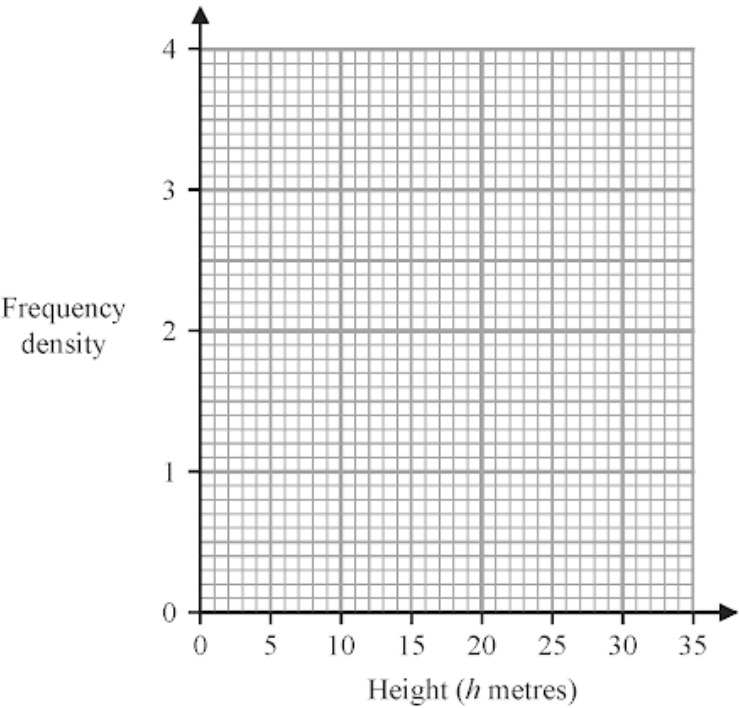
Show clear algebraic working.

(5)

The table gives information about the heights, in metres, of the trees in a park.

Height (h metres)	Frequency
$0 < h \leq 2$	5
$2 < h \leq 5$	12
$5 < h \leq 10$	18
$10 < h \leq 20$	14
$20 < h \leq 35$	9

On the grid, draw a histogram for this information.



(3)

7. 4MA1/1HR_Summer_2024_Q10

Here are the numbers of goals scored by a hockey team in its 11 games this season.

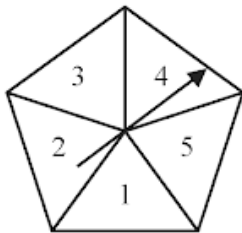
0 1 2 2 3 4 4 6 7 9 11

Work out the interquartile range of the numbers of goals.

(2)

8. 4MA1/1HR_Summer_2024_Q2

Here is a biased spinner.



The table gives information about the probability that, when the spinner is spun once, it will land on each number.

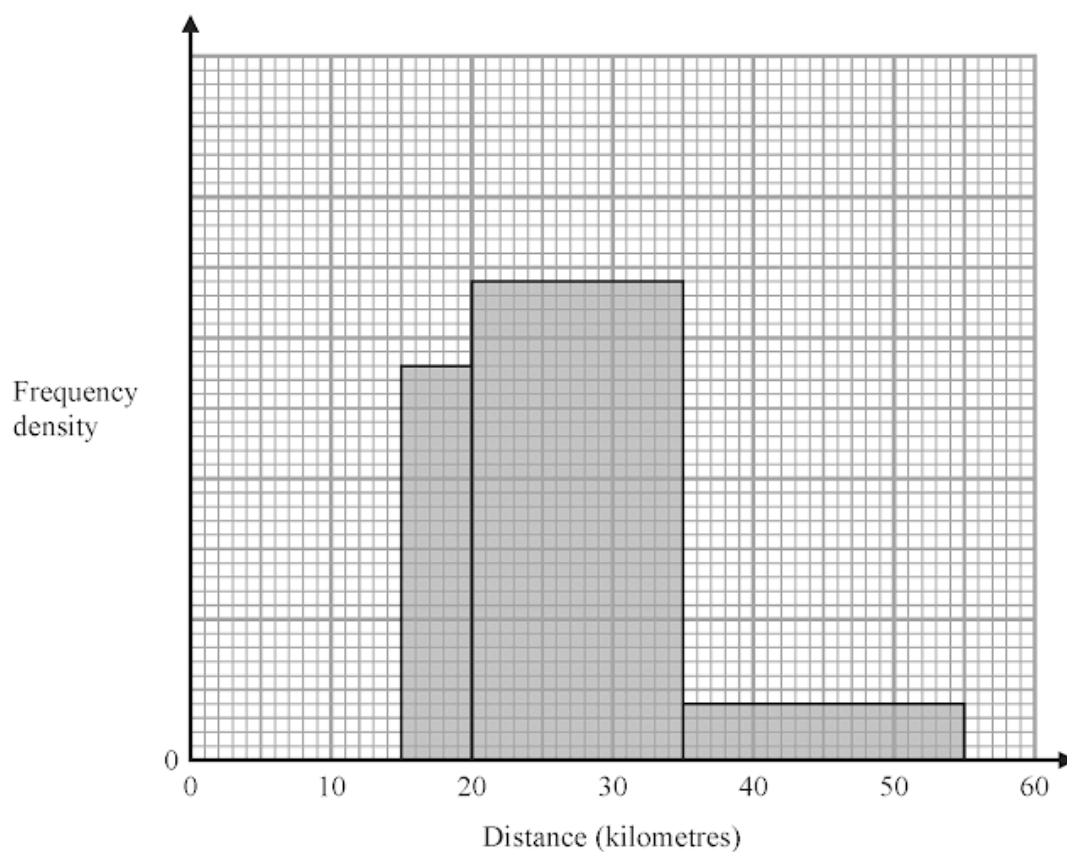
Number	1	2	3	4	5
Probability	$2x$	0.27	0.04	x	0.12

Alexis is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on an odd number.

(4)

The incomplete histogram shows some information about the distances, in kilometres, that 100 adults ran last week.



All of the adults ran at least 5 kilometres.

None of the adults ran more than 55 kilometres.

14 adults ran between 15 kilometres and 20 kilometres.

Complete the histogram.

(3)

There are 20 sweets in a box.

15 of the sweets are red

5 of the sweets are yellow

Fred takes at random 3 sweets from the box.

Work out the probability that Fred takes at least one sweet of each colour from the box.

(4)

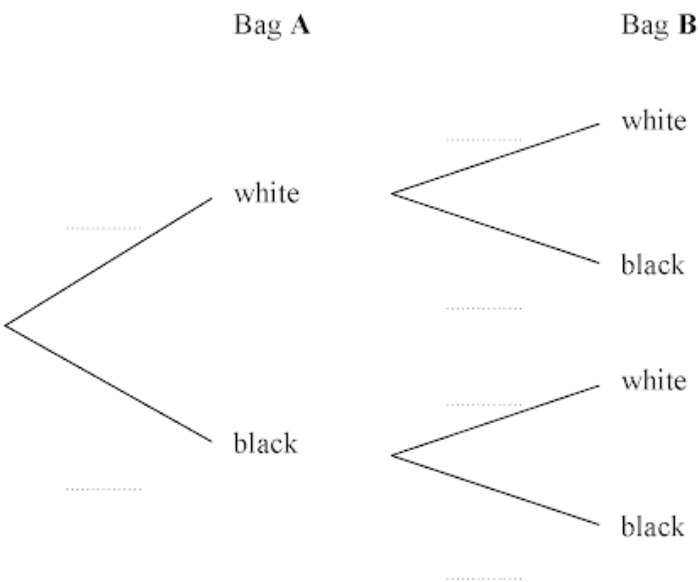
Harman has two bags of beads.

In bag **A**, there are 3 white beads and 7 black beads.

In bag **B**, there are 5 white beads and 4 black beads.

Harman takes at random a bead from bag **A** and a bead from bag **B**

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Harman takes two beads of the same colour.

(3)

$\mathcal{E} = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$
 $A = \{\text{even numbers}\}$
 $B = \{23, 29, 31\}$
 $C = \{\text{multiples of } 3\}$

(a) List the members of the set

(i) $B \cup C$ (1)

(ii) $A' \cap C$ (1)

(b) Is it true that $B \cap C = \emptyset$?

Tick (✓) one of the boxes below.

Yes	No
☐	☐

Give a reason for your answer. (1)

The set D has 4 members and is such that $D \cap (A \cup C) = \emptyset$

(c) List the members of set D (2)

450 students were asked how they travelled to school on Monday.
Each student walked or travelled by bus or travelled by car or travelled by bicycle.
Each student used just one method of travel.

One of these students is chosen at random.
The table shows information about the probability of each method of travel.

Method of travel	walk	bus	car	bicycle
Probability	0.20	x	$2x$	0.26

Work out how many of the 450 students travelled by car.

(4)

The table shows information about the times, in minutes, that 80 patients had to wait to see a doctor.

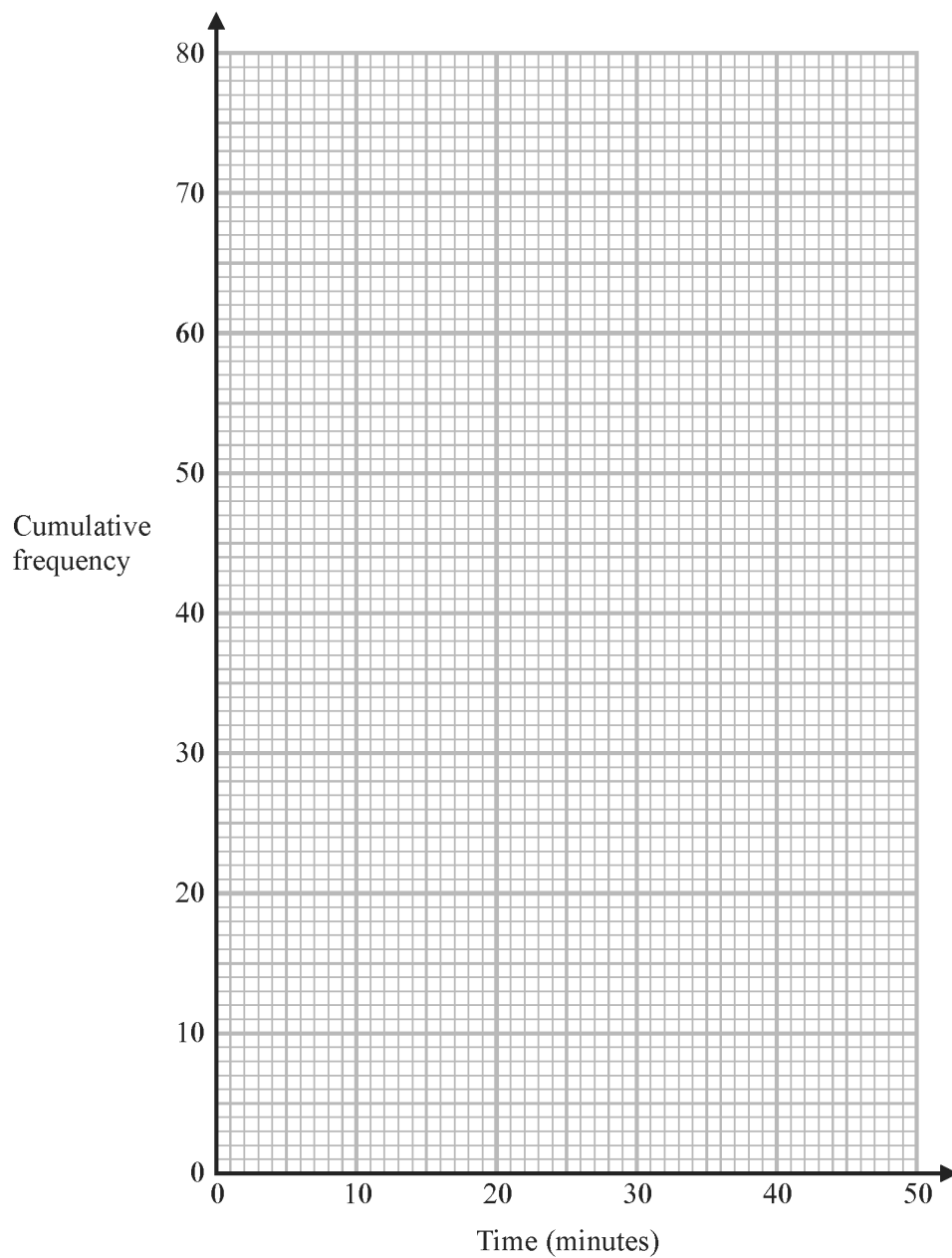
Time (W minutes)	Frequency
$0 < W \leq 10$	7
$10 < W \leq 20$	10
$20 < W \leq 30$	15
$30 < W \leq 40$	32
$40 < W \leq 50$	16

(a) Complete the cumulative frequency table below.

Time (W minutes)	Cumulative frequency
$0 < W \leq 10$	
$0 < W \leq 20$	
$0 < W \leq 30$	
$0 < W \leq 40$	
$0 < W \leq 50$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the median.

..... minutes

(1)

(d) Use your graph to find an estimate for the interquartile range.

..... minutes

(2)

Here is a list of six numbers written in order of size.

x 5 y z 10 12

The numbers have

- a range of 9
- a median of 8
- a mode of 10

Find the value of x , the value of y and the value of z

$x =$

$y =$

$z =$

3 marks

The table shows information about the frame size, in cm, of 60 bicycles sold in a shop.

Frame size (S cm)	Frequency
$30 < S \leq 36$	4
$36 < S \leq 42$	14
$42 < S \leq 48$	18
$48 < S \leq 54$	19
$54 < S \leq 60$	5

(a) Write down the modal class.

.....
(1)

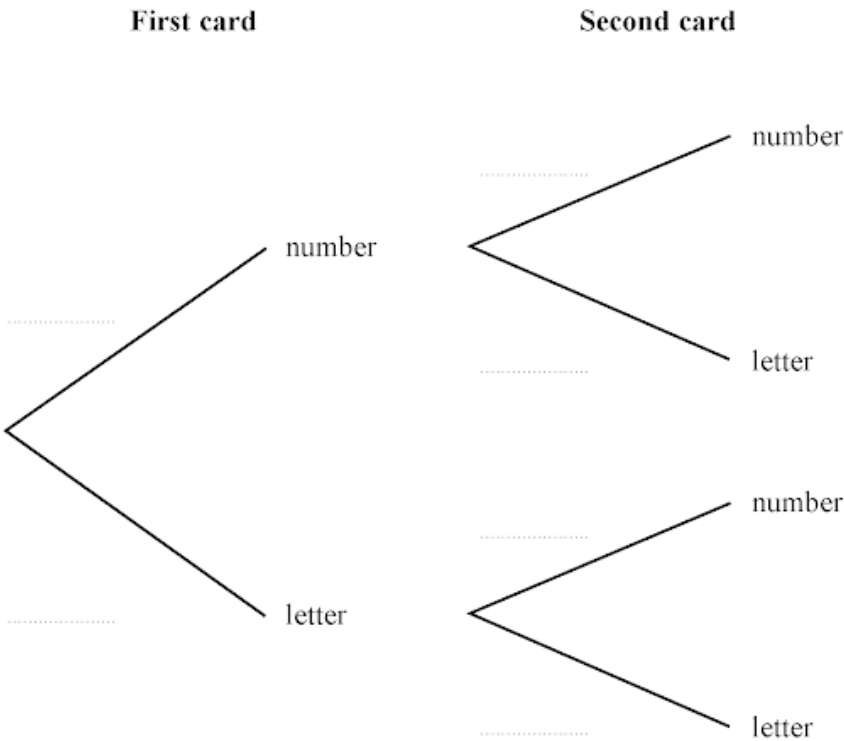
(b) Work out an estimate for the mean frame size.

..... cm
(4)

Here are 9 cards. Each card has either a number on it or a letter on it.



- Tomas is playing a game.
- Tomas takes at random one of the cards and keeps it.
- Tomas then takes at random another card and keeps it.
- (a) Complete the probability tree diagram.



(2)

(b) Work out the probability that each of the two cards has a number on it.

(2)

(c) Work out the probability that there will be one card with a number on it and one card with a letter on it.

(3)

Some members of a library were asked to name the type of book that they each liked to read the best.

One of the members is chosen at random.

The table shows information about the probability of the type of book that this member answered.

Type of book	comedy	romance	mystery	thriller
Probability	0.24	0.40	$3x$	x

48 members answered comedy books.

Work out how many of the members answered mystery books.

.....

4 marks

80 students entered a dancing competition.

The table gives information about the length of time, in minutes, for which each student spent dancing.

Time (m)	Frequency
$0 < m \leq 12$	11
$12 < m \leq 24$	25
$24 < m \leq 36$	23
$36 < m \leq 48$	15
$48 < m \leq 60$	6

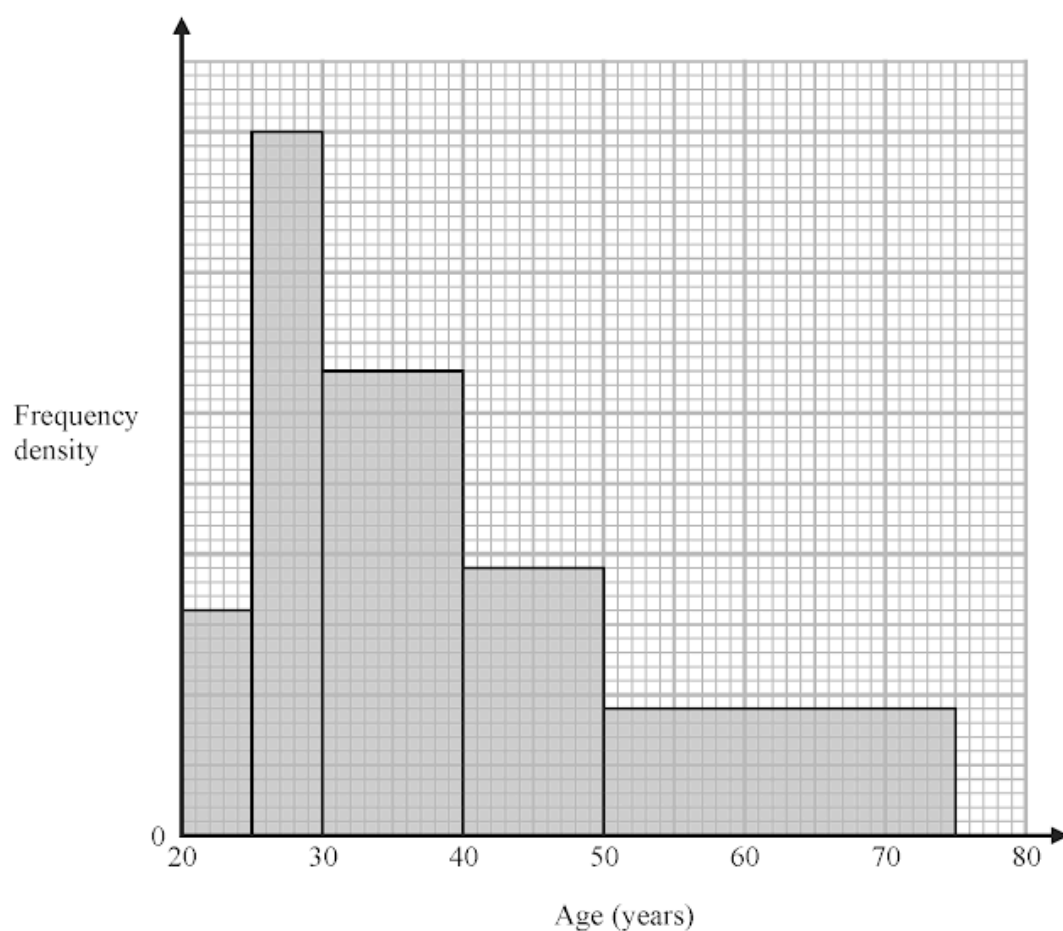
Work out an estimate for the mean length of time the students spent dancing.

..... minutes

4 marks

Some people attend a concert.

The histogram shows information about the ages of these people.



Work out an estimate for the percentage of these people who are aged more than 55 years.

Give your answer correct to one decimal place.

The table gives information about the times taken by 80 people to run a race.

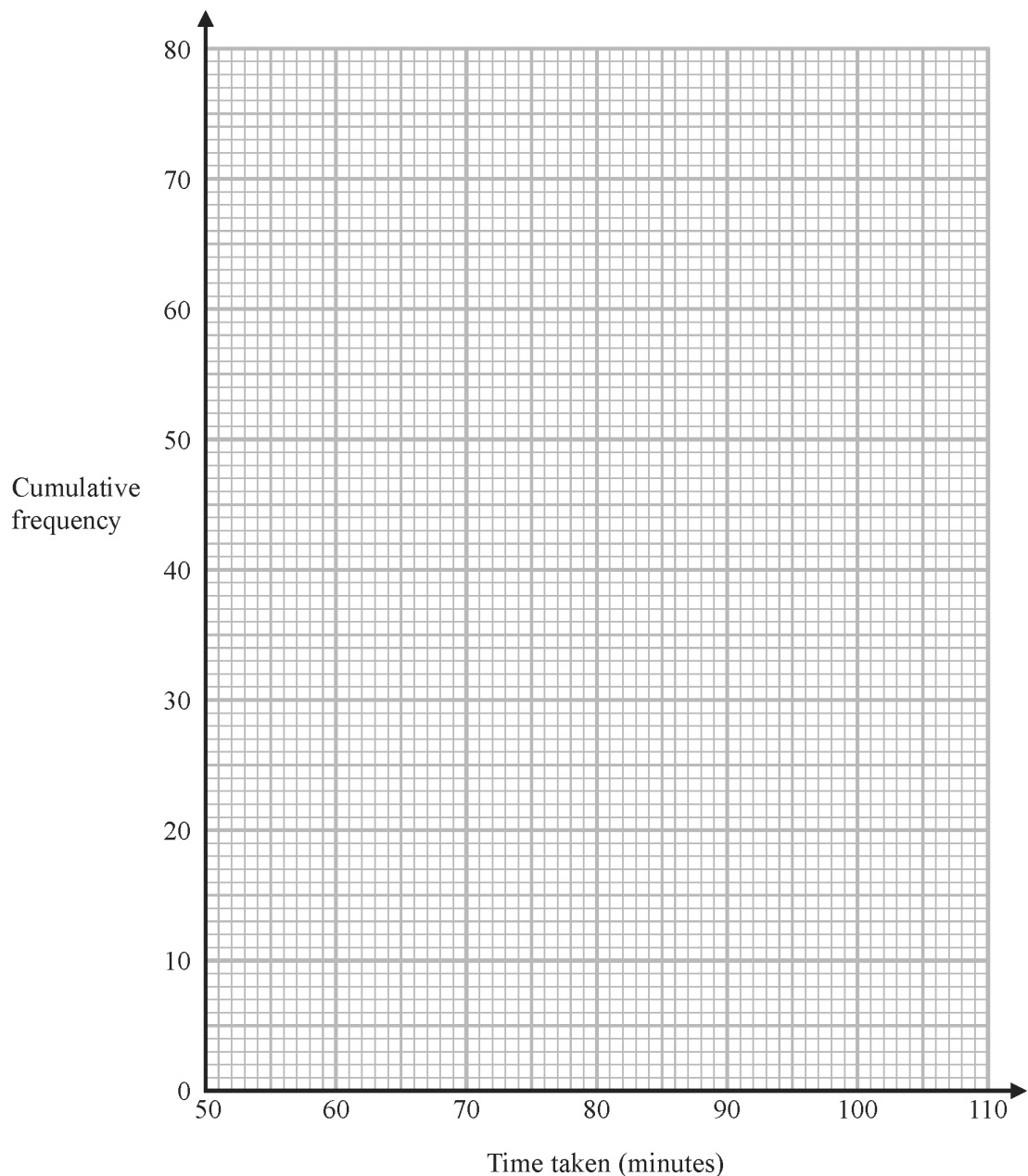
Time taken (t minutes)	Frequency
$50 < t \leq 60$	15
$60 < t \leq 70$	16
$70 < t \leq 80$	21
$80 < t \leq 90$	14
$90 < t \leq 100$	8
$100 < t \leq 110$	6

(a) Complete the cumulative frequency table.

Time taken (t minutes)	Cumulative frequency
$50 < t \leq 60$	
$50 < t \leq 70$	
$50 < t \leq 80$	
$50 < t \leq 90$	
$50 < t \leq 100$	
$50 < t \leq 110$	

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the median time taken.

..... minutes

(1)

(d) Use your graph to find an estimate for the interquartile range of the times taken.

..... minutes

(2)

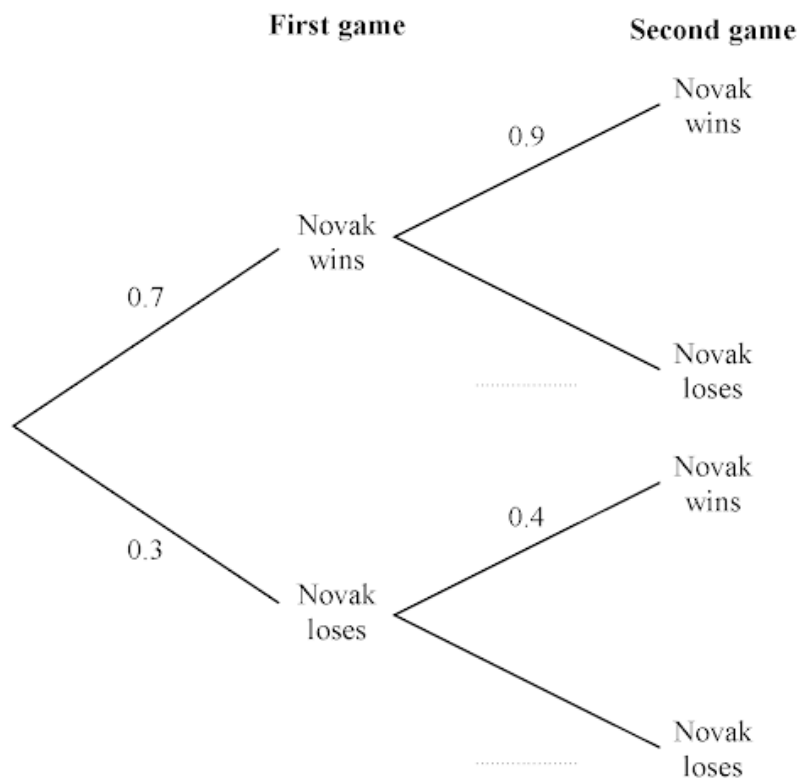
Novak is going to play two games of tennis.

The probability that he wins the first game is 0.7

If he wins the first game, the probability that he wins the second game is 0.9

If he loses the first game, the probability that he wins the second game is 0.4

(a) Complete the probability tree diagram.



(1)

(b) Work out the probability that Novak wins both games of tennis.

(2)

Moeen has a biased 6-sided dice.

The table gives information about the probability that, when the dice is thrown, it will land on each number.

Number	1	2	3	4	5	6
Probability	x	0.15	0.5	y	0.13	0.03

(a) Show that $x + y = 0.19$

(2)

Given that $3x - y = 0.09$

and $x + y = 0.19$

(b) work out the value of x and the value of y
Show clear algebraic working.

$x = \dots\dots\dots$

$y = \dots\dots\dots$

(3)

24. 4MA1/1HR_Summer_2023_Q5

Each time Evie plays a game against her computer, she will win or lose.

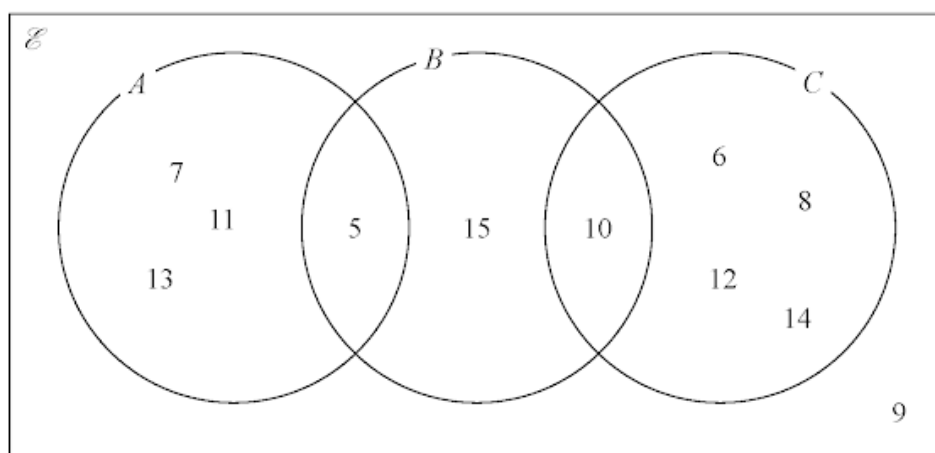
For each game, the probability that Evie will win is 0.74

Evie is going to play 300 games against her computer.

Work out an estimate for the number of games that Evie will lose.

.....
2 marks

Here is a Venn diagram.



(a) Write down the numbers that are in the set

(i) A

(1)

(ii) $B \cup C$

(1)

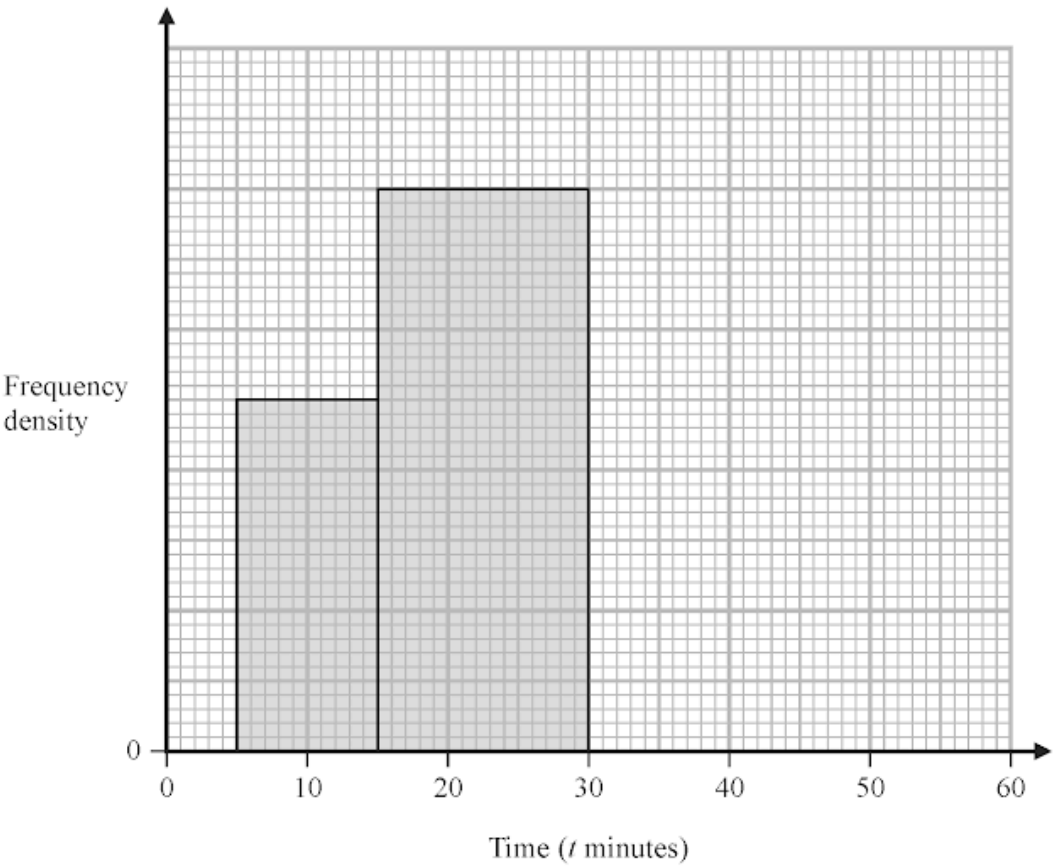
Dominic writes down $9 \notin C$

(b) Explain why Dominic is correct.

(1)

The incomplete table and incomplete histogram give information about the times, in minutes, that 140 people waited at a station for a train.

Time (t minutes)	Frequency
$0 < t \leq 5$	23
$5 < t \leq 15$	
$15 < t \leq 30$	
$30 < t \leq 40$	18
$40 < t \leq 60$	14

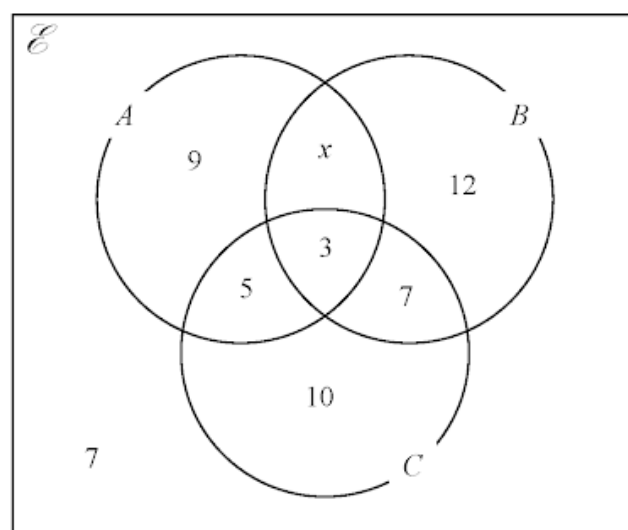


Complete the table and the histogram.

4 marks

The Venn diagram shows a universal set $\mathcal{E}$ and sets A , B and C

The numbers and the letter x represent **numbers** of elements.



Given that $n(A \cup B) = 42$

(a) find the value of x

$x =$
(1)

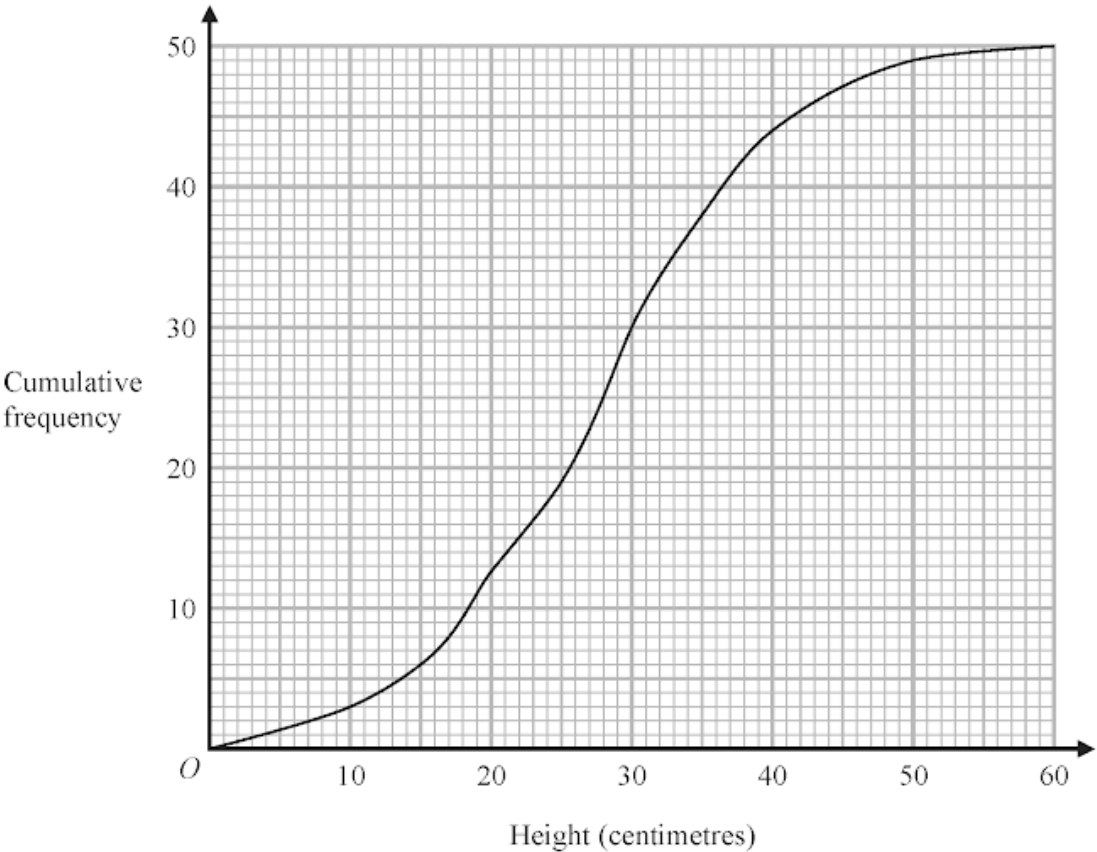
(b) Find $n(A')$

.....
(1)

(c) Find $n(B' \cap C)$

.....
(1)

The cumulative frequency graph shows information about the heights, in centimetres, of 50 plants in a flowerbed.



(a) Use the graph to find an estimate for the median height of these plants.

..... centimetres
(1)

(b) Use the graph to find the frequency for the class interval $30 < \text{Height} \leq 40$

.....
(1)

(c) Use the graph to find an estimate for the number of plants with a height greater than 35 centimetres.

.....
(2)

29. 4MA1/1H_Summer_2023_Q10

Here are the test marks of 15 students.

7 10 14 15 16 17 18 19 20 20 23 25 30 39 40

Find the interquartile range of these marks.

.....
2 marks

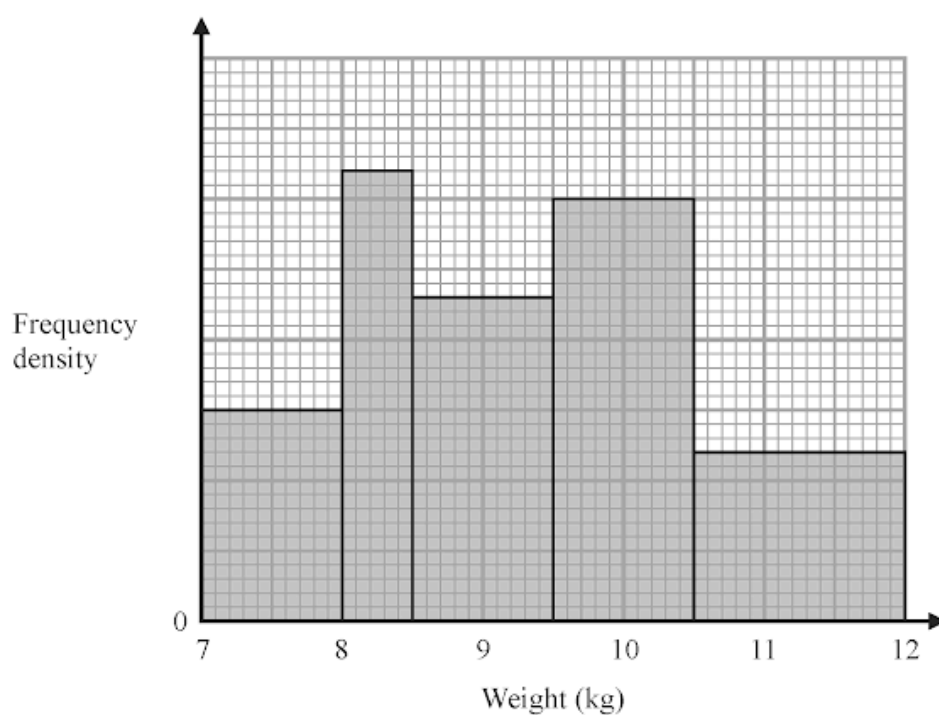
The table gives information about the number of minutes that Abby spent walking each day in September.

Number of minutes (M)	Frequency
$0 < M \leq 30$	5
$30 < M \leq 60$	6
$60 < M \leq 90$	8
$90 < M \leq 120$	9
$120 < M \leq 150$	2

Work out an estimate for the total number of minutes that Abby spent walking in September.

..... minutes

3 marks



The histogram gives information about the weights, in kg, of all the watermelons in a field.

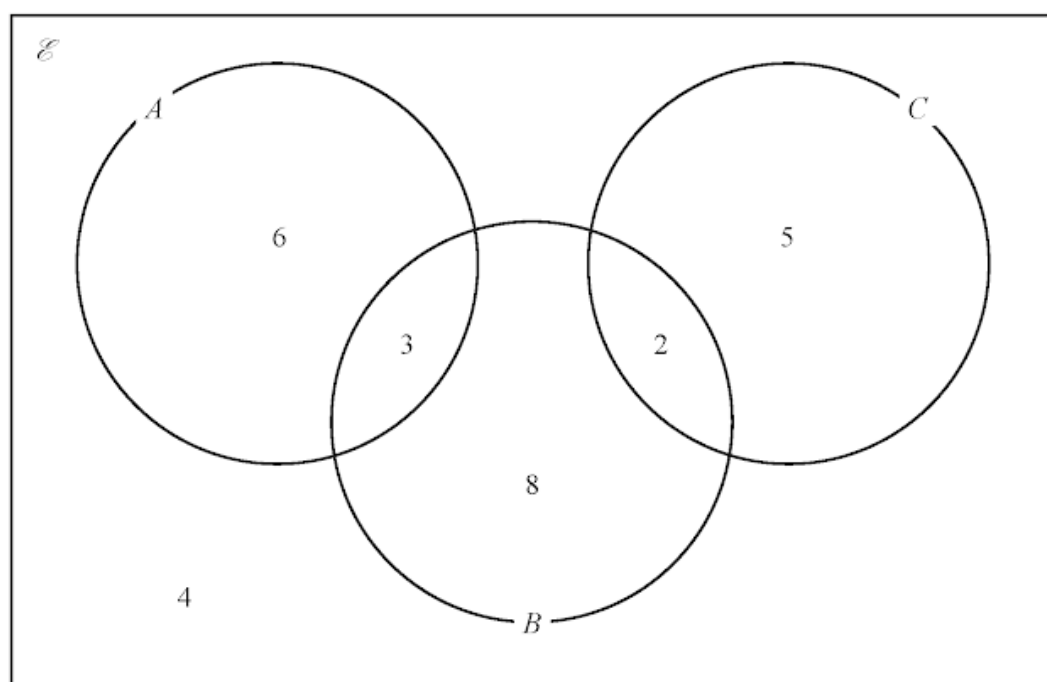
There are 16 watermelons with a weight between 8 kg and 8.5 kg

Work out the total number of watermelons in the field.

.....

3 marks

The Venn diagram shows a universal set $\mathcal{E}$ and three sets A , B and C .



6, 3, 8, 2, 5 and 4 represent the **numbers** of elements.

Find

(i) $n(A \cup B)$

.....
(1)

(ii) $n(A \cap C)$

.....
(1)

(iii) $n(B \cap C')$

.....
(1)

(iv) $n(A' \cup B' \cup C')$

.....
(1)

Hector has a bag that contains 12 counters.

There are 7 green counters and 5 red counters in the bag.

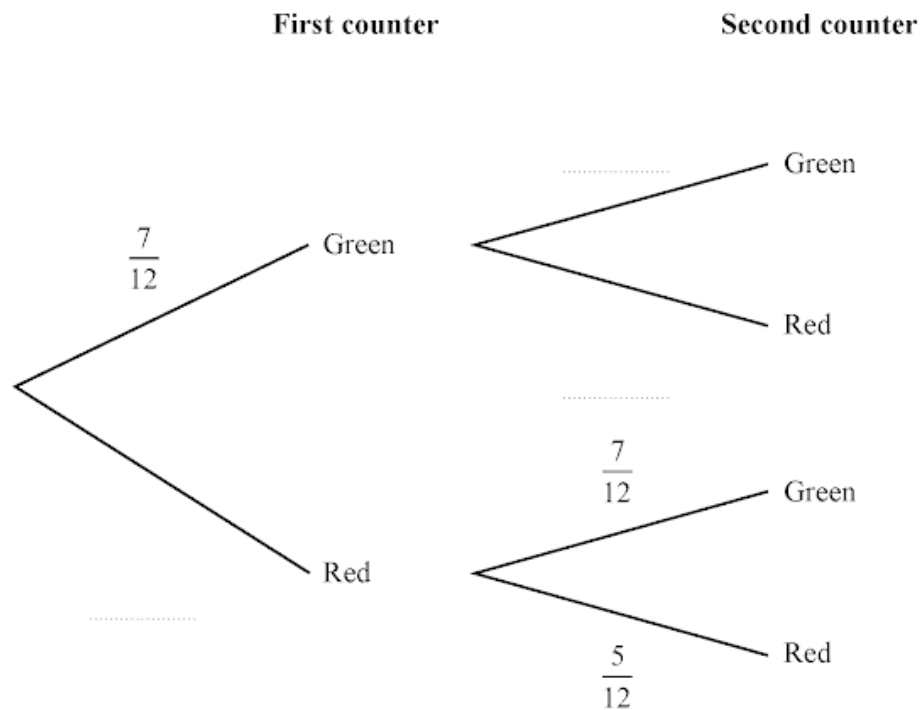
Hector takes at random a counter from the bag.

He looks at the counter and puts the counter back into the bag.

Hector then takes at random a second counter from the bag.

He looks at the counter and puts the counter back into the bag.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that both counters are red.

.....
(2)

Meghan has a jar containing 15 counters.

There are only blue counters, green counters and red counters in the jar.

Hector is going to take at random one of the counters from his bag of 12 counters.

He will look at the counter and put the counter back into the bag.

Hector is then going to take at random a second counter from his bag.

He will look at the counter and put the counter back into the bag.

Meghan is then going to take at random one of the counters from her jar of counters.

She will look at the counter and put the counter back into the jar.

The probability that the 3 counters each have a different colour is $\frac{7}{24}$

(c) Work out how many blue counters there are in the jar.

34. 4MA1/1HR_Winter_2022_Q5

Yusuf sat 8 examinations.

Here are his marks for 5 of the examinations.

68 72 75 77 80

For his results in all 8 examinations

the mode of his marks is 80

the median of his marks is 74

the range of his marks is 16

Find Yusuf's marks for each of the other 3 examinations.

.....

.....

.....

4 marks

35. 4MA1/1HR_Winter_2022_Q2

Each time John plays a game, the probability that he wins the game is 0.65

John is going to play the game 300 times.

Work out an estimate for the number of games that John wins.

.....
2 marks

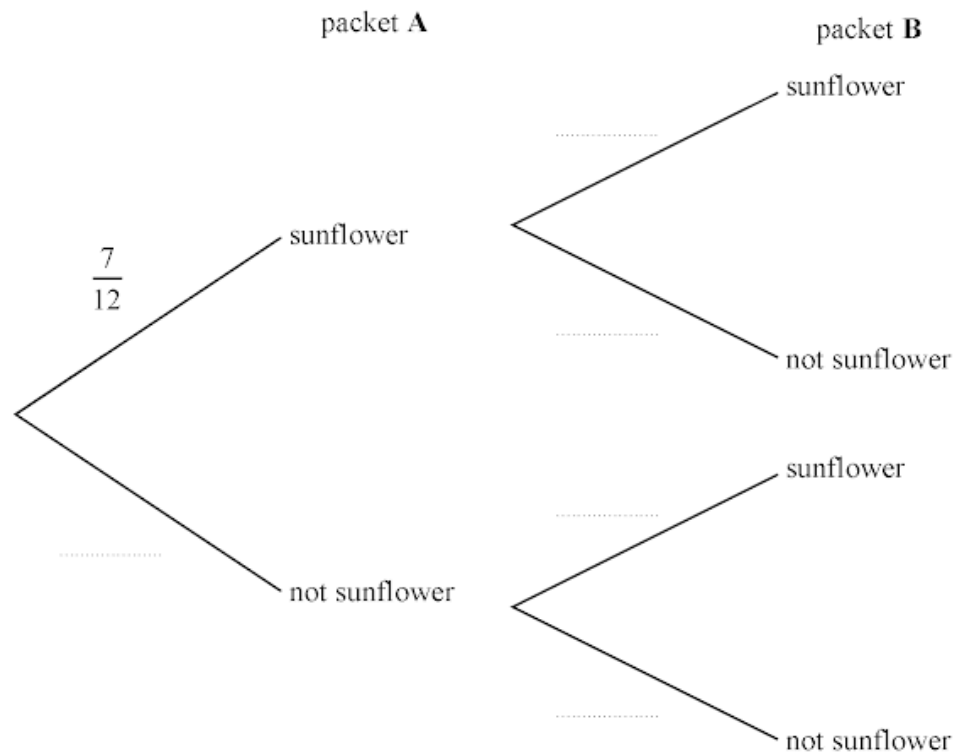
Aika has 2 packets of seeds, packet **A** and packet **B**

There are 12 seeds in packet **A** and 7 of these are sunflower seeds.

There are 15 seeds in packet **B** and 8 of these are sunflower seeds.

Aika is going to take at random a seed from packet **A** and a seed from packet **B**

(a) Complete the probability tree diagram.



(2)

(b) Calculate the probability that Aika will take two sunflower seeds.

(2)

The table gives information about the times, in minutes, taken by 80 customers to do their shopping in a supermarket.

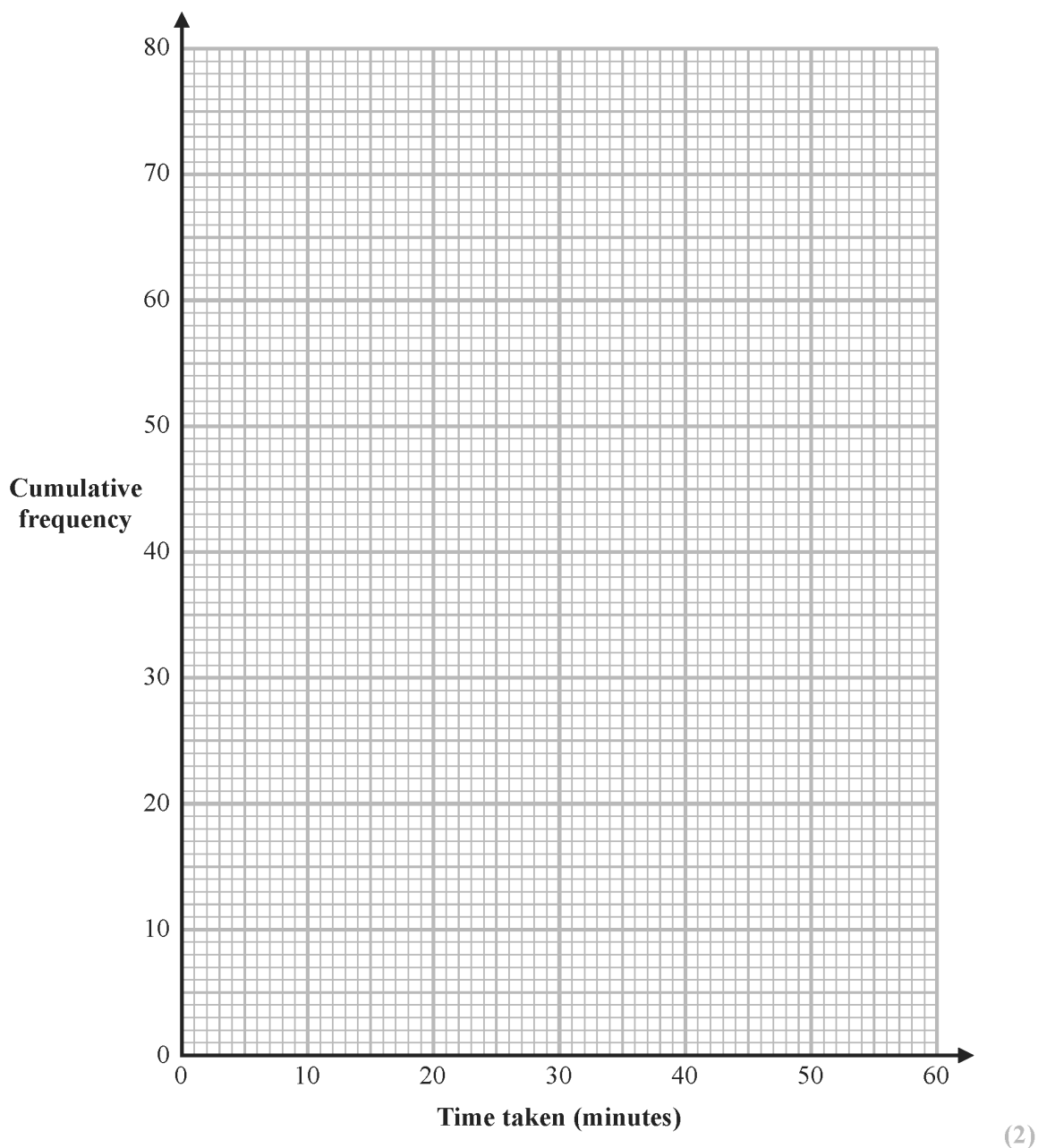
Time taken (t minutes)	Frequency
$0 < t \leq 10$	7
$10 < t \leq 20$	26
$20 < t \leq 30$	24
$30 < t \leq 40$	14
$40 < t \leq 50$	7
$50 < t \leq 60$	2

(a) Complete the cumulative frequency table.

Time taken (t minutes)	Cumulative frequency
$0 < t \leq 10$	
$0 < t \leq 20$	
$0 < t \leq 30$	
$0 < t \leq 40$	
$0 < t \leq 50$	
$0 < t \leq 60$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.



(c) Use your graph to find an estimate for the median time taken.

..... minutes
(1)

One of the 80 customers is chosen at random.

(d) Use your graph to find an estimate for the probability that the time taken by this customer was more than 42 minutes.

.....
(2)

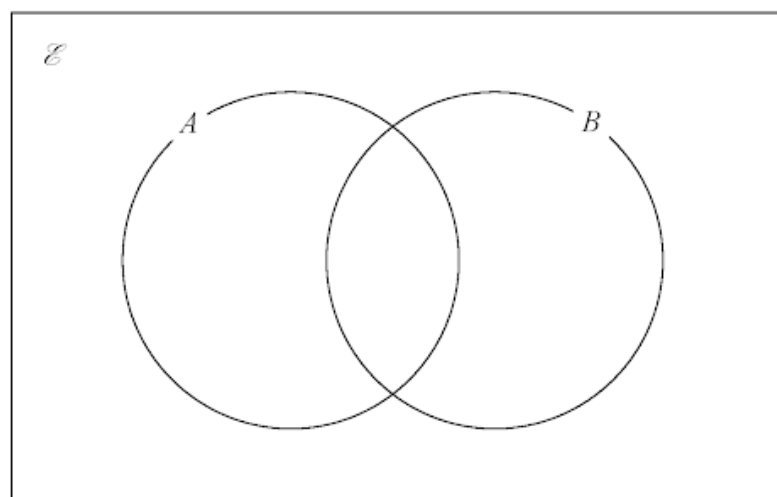
$$\mathcal{E} = \{4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$$

$$A \cap B = \{5, 10, 15\}$$

$$B' = \{7, 8, 9, 11, 12, 13, 14\}$$

$$A' = \{4, 6, 7, 8, 14\}$$

Complete the Venn diagram for this information.



3 marks

Jenny has six cards.

Each card has a whole number written on it so that

- the smallest number is 5
- the largest number is 24
- the median of the six numbers is 14
- the mode of the six numbers is 8

Jenny arranges her cards so that the numbers are in order of size.

5					24
---	-------	-------	-------	-------	----

- (a) For the remaining four cards, write on each dotted line a number that could be on the card.

(3)

A basketball team plays 6 games.

After playing 5 games, the team has a mean score of 21 points per game.

After playing 6 games, the team has a mean score of 23 points per game.

- (b) Work out the number of points the team scored in its 6th game.

.....
(3)

Elliot has x counters.

Each counter has one red face and one green face.

Elliot spreads all the counters out on a table and sees that the number of counters showing a red face is 5

Elliot then picks at random one of the counters and turns the counter over.
He then picks at random a second counter and turns the counter over.

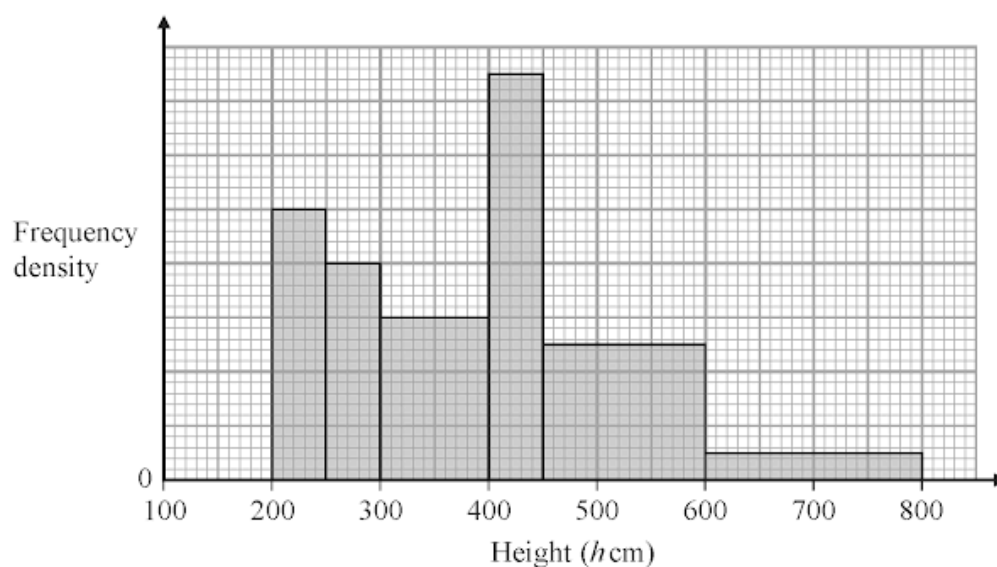
The probability that there are still 5 counters showing a red face is $\frac{19}{32}$

Work out the value of x
Show clear algebraic working.

$x =$

5 marks

The histogram gives information about the height, h cm, of each tree in part of a forest.



There are no trees for which $h \leq 200$ and for which $h > 800$

The number of trees for which $300 < h \leq 400$ is 8 fewer than the number of trees for which $400 < h \leq 500$

Work out an estimate for the number of trees in this part of the forest that have a height greater than 500 cm.

3 marks

42. 4MA1/1HR_Summer_2022_Q13

Here is the number of runs scored by a baseball team in each of its 15 games this season.

The number of runs have been arranged in order of size.

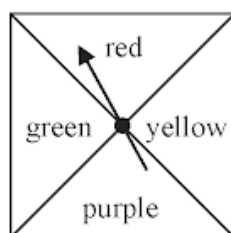
0 1 1 3 5 6 7 7 8 9 9 12 12 15 16

Work out the interquartile range of the number of runs.

.....

2 marks

Here is a biased spinner.



When the spinner is spun once, the probabilities that it lands on red or on yellow or on green are given in the table.

Colour	red	yellow	purple	green
Probability	0.25	0.2		0.2

(a) Work out the probability that the spinner lands on red or on yellow.

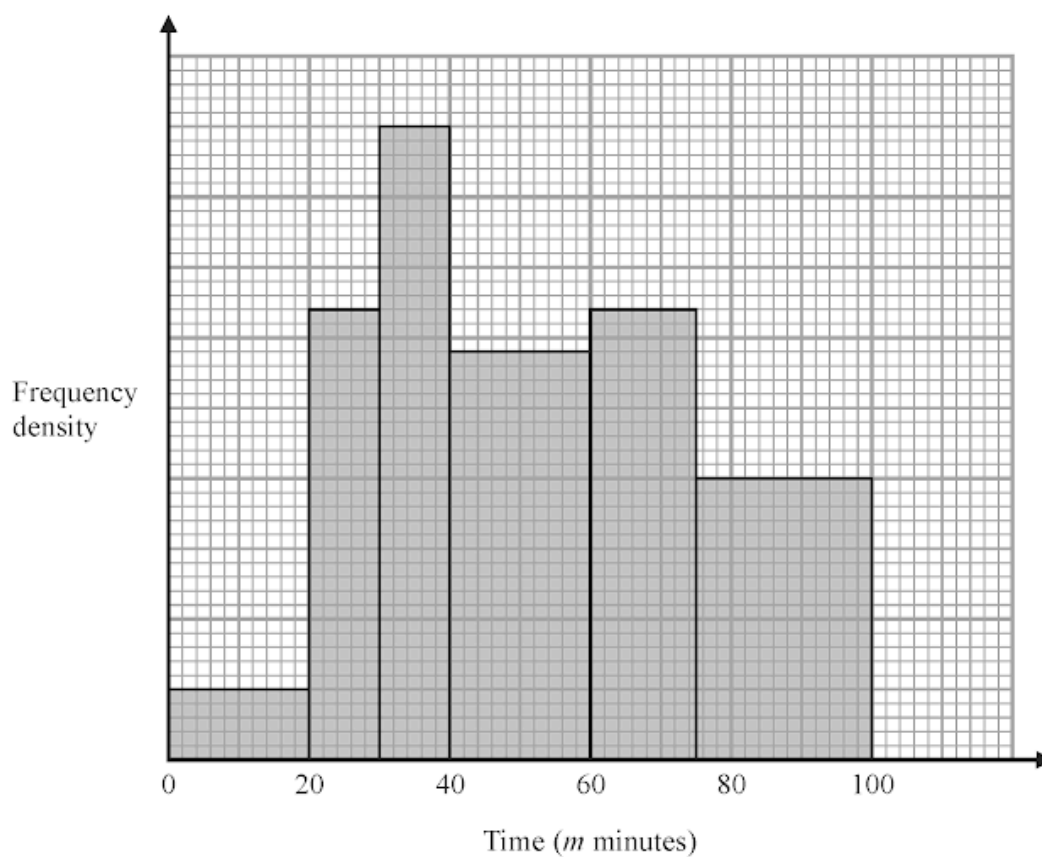
.....
(1)

Yang is going to spin the spinner 300 times.

(b) Work out an estimate for the number of times the spinner will land on purple.

.....
(3)

The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$

There are 10 children for whom $m \leq 20$

Work out an estimate for the number of children for whom $50 < m \leq 80$

3 marks

45. 4MA1/1H_Summer_2022_Q15

Abraham is going to play a computer game.
Abraham can win the game, draw the game or lose the game.

For any game that Abraham plays

the probability that he wins the game is 0.3
the probability that he draws the game is 0.5
the probability that he loses the game is 0.2

When Abraham wins a game, he scores +10 points.

When Abraham draws a game, he scores 0 points.

When Abraham loses a game, he scores -5 points.

Abraham plays 3 games and the points he scores in each of the 3 games are added together to get his total score.

Work out the probability that when he has played 3 games his total score is 0 points.

.....
4 marks

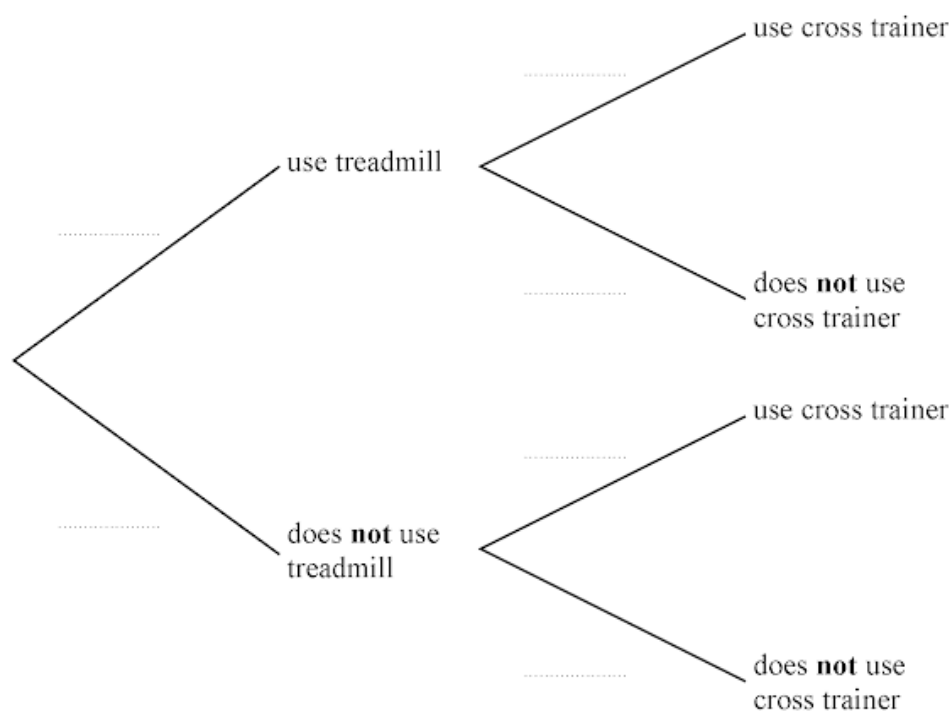
Rudolf goes to the gym.

The probability that he will use the treadmill is 0.8

When he uses the treadmill, the probability that he will use the cross trainer is 0.3

When he does **not** use the treadmill, the probability that he will use the cross trainer is 0.6

(a) Complete the probability tree diagram for this information.

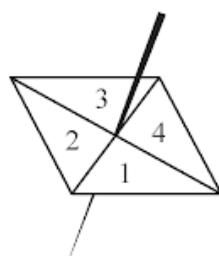


(2)

(b) Work out the probability that Rudolf uses both the treadmill and the cross trainer.

(2)

Here is a biased 4-sided spinner.



The table gives the probabilities that, when the spinner is spun once, it will land on 1 or it will land on 3

Number	1	2	3	4
Probability	0.26		0.18	

The probability that the spinner will land on 2 is equal to the probability that the spinner will land on 4

Ravina is going to spin the spinner a number of times.

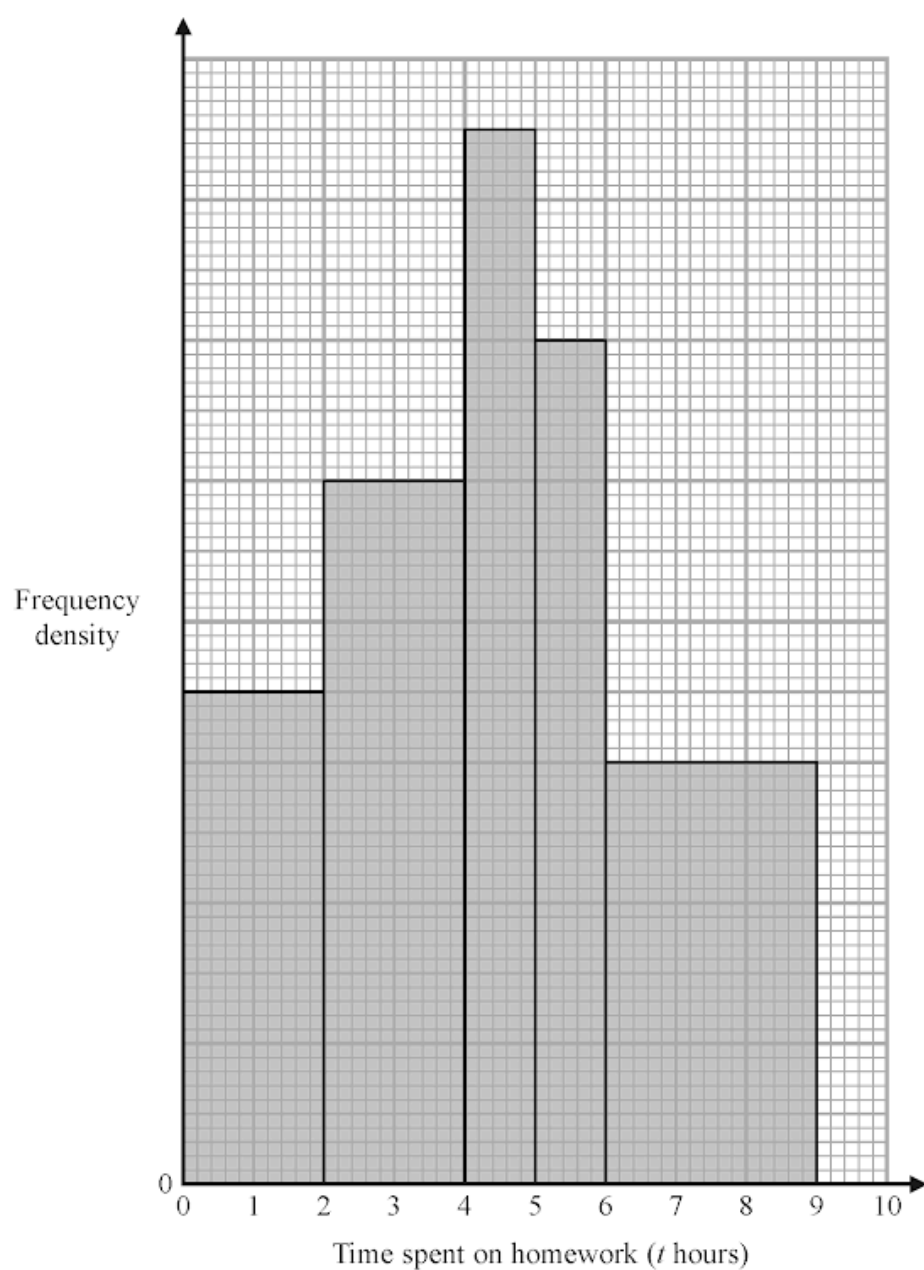
Ravina works out that an estimate for the number of times the spinner will land on 3 is 45

Work out an estimate for the number of times the spinner will land on 4

.....

4 marks

The histogram and the table give some information about the amounts of time, in hours, that Year 11 students at Bergdesh Academy spent, in total, on their homework last week. No student in Year 11 spent longer than 9 hours on their homework.



Time spent on homework (t hours)	Frequency
$0 < t \leq 2$	28
$2 < t \leq 4$	
$4 < t \leq 5$	
$5 < t \leq 6$	
$6 < t \leq 9$	

Using the information in the histogram and in the table, work out an estimate for the mean amount of time the Year 11 students spent on their homework last week.
Give your answer in hours correct to 3 significant figures.

..... hours

(Total for Question is 5 marks)

The manager of a call centre asked the 120 people, who rang the call centre last week, how long they each waited before their call was answered.

The table gives information about their replies.

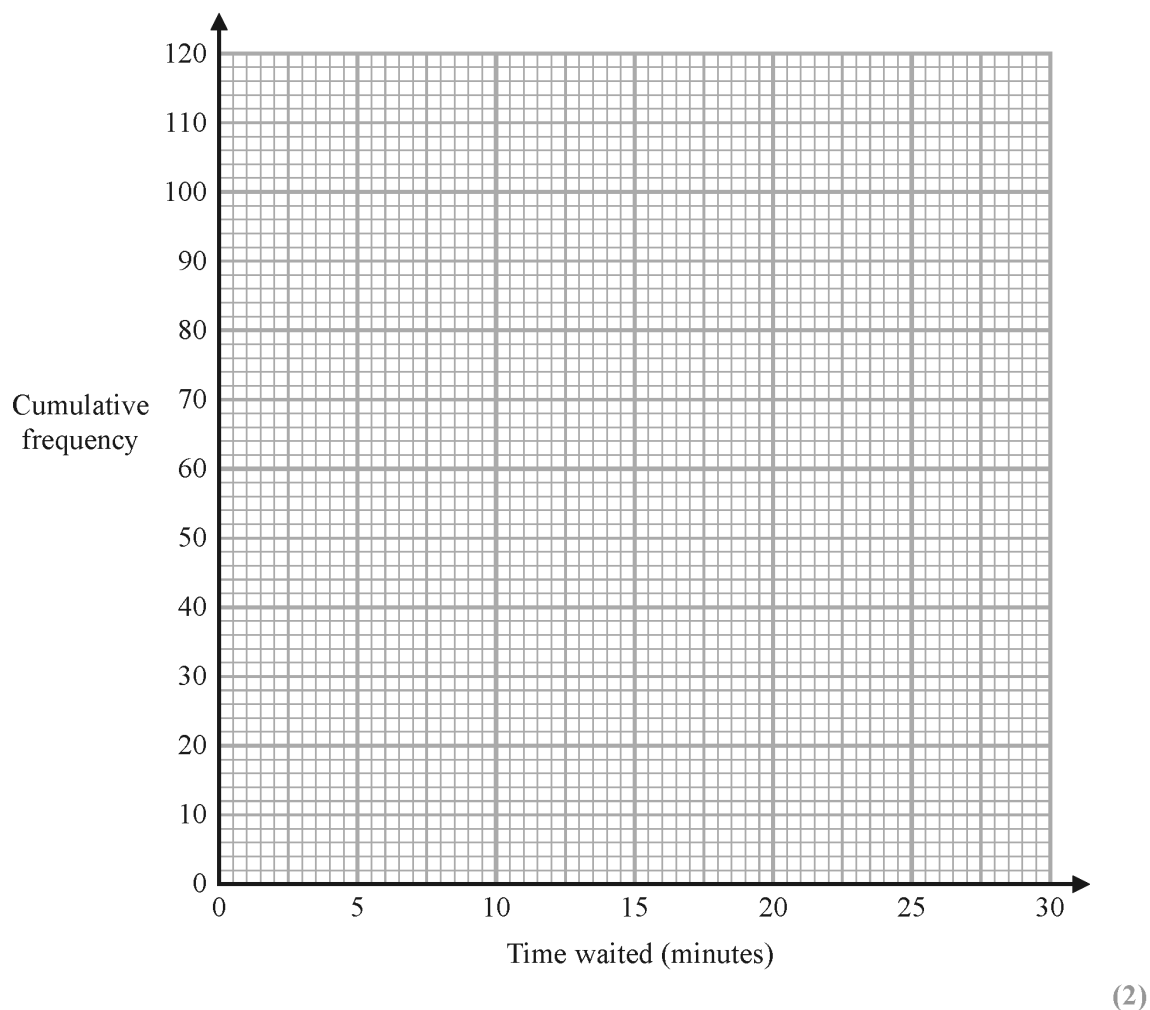
Time waited (t minutes)	Frequency
$0 < t \leq 5$	8
$5 < t \leq 10$	15
$10 < t \leq 15$	17
$15 < t \leq 20$	28
$20 < t \leq 25$	33
$25 < t \leq 30$	19

(a) Complete the cumulative frequency table.

Time waited (t minutes)	Cumulative frequency
$0 < t \leq 5$	
$0 < t \leq 10$	
$0 < t \leq 15$	
$0 < t \leq 20$	
$0 < t \leq 25$	
$0 < t \leq 30$	

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(c) Use your graph to find an estimate for the median of the times waited.

..... minutes
(1)

(d) Using your graph, find an estimate for the percentage of the 120 people who said that they waited longer than 23 minutes before their call was answered.
Show your working clearly.

..... %
(2)